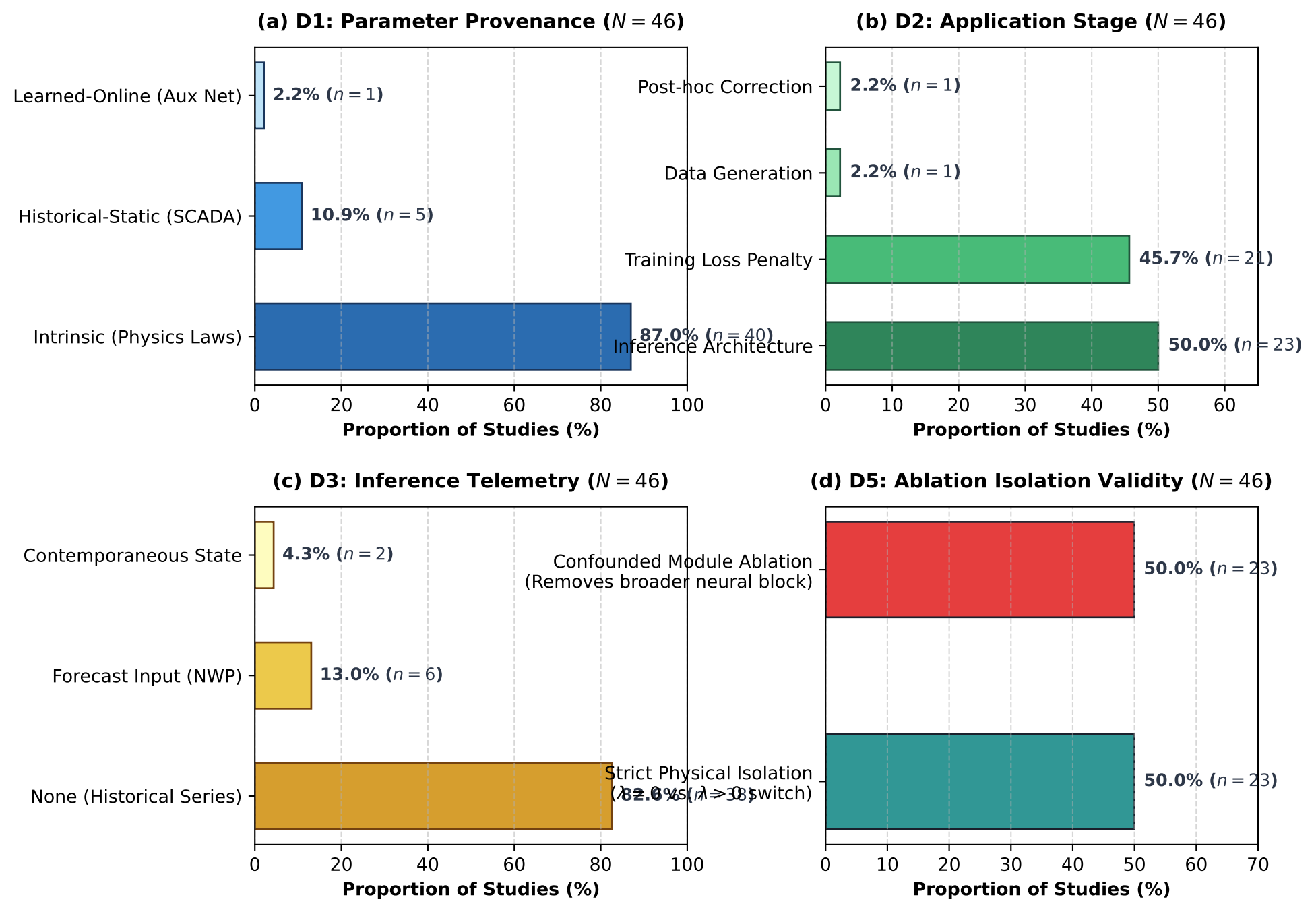
(a) D1: Parameter Provenance (N = 46)
Learned-Online (Aux Net)
2.2% (n = 1)
Historical-Static (SCADA)
10.9% (n = 5)
Intrinsic (Physics Laws)
87.0% (n = 40)
0
20
40
60
80
100
Proportion of Studies (%)
(b) D2: Application Stage (N = 46)
Post-hoc Correction
2.2% (n = 1)
Data Generation
2.2% (n = 1)
Training Loss Penalty
45.7% (n = 21)
Inference Architecture
50.0% (n = 23)
0
10
20
30
40
50
60
Proportion of Studies (%)
(c) D3: Inference Telemetry (N = 46)
Contemporaneous State
4.3% (n = 2)
Forecast Input (NWP)
13.0% (n = 6)
None (Historical Series)
82.6% (n = 38)
0
20
40
60
80
100
Proportion of Studies (%)
(d) D5: Ablation Isolation Validity (N = 46)
Confounded Module Ablation
(Removes broader neural block)
50.0% (n = 23)
Strict Physical Isolation
(Gate Switch)
50.0% (n = 23)
0
10
20
30
40
50
60
70
Proportion of Studies (%)